Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/03/2025
PI Name:	Robertson, Ian Ph.D.		
Project Title:	AR/VR and Hybrid Reality for Verification Assessments: Phase I		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/12/2023	**End Date:**	07/12/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Extended Reality (XR) technologies may provide significant cost savings in that they require little to no hardware, construction costs, or storage facilities. It is hypothetically feasible for XR methods to faithfully replicate NASA systems (e.g., a high-fidelity virtual reality simulation can approximate the appearance, size, and functionality of a real-world experience), and there is a body of literature supporting the use of XR methods, especially virtual reality, for design evaluations. Currently, it is unclear how suitable XR methods are for use in verification activities. The biggest concern is that verification data collected with XR technology would not be comparable to data collected in a physical mockup using traditional methods, resulting in additional risk when used for flight readiness decisions. Therefore, NASA's current position is to allow the use of XR technologies for developmental test activities only. If providers are to employ XR technology for verification in the future, NASA needs data and theoretical knowledge to develop guidelines for how XR technology can be applied without increasing human health and performance risks. The proposed work will help NASA better understand the strengths and limitations of XR technologies, and will identify research gaps, which		

	may help foster innovation and focus research efforts.
Rationale for HRP Directed Research:	This proposal qualifies for directed study due to the constrained nature of the work. To understand this issue, investigators will need to be able to directly communicate with NASA subject matter experts (SMEs) who work with spaceflight programs using XR for some activities that could be proprietary. External researchers would not be able to easily access this information and thus directed research would be more cost effective. Another factor is that commercial providers are currently pushing to employ XR technologies for verification activities, leaving insufficient time for solicitation.
Research Impact/Earth Benefits:	The data from Phase 1 will inform future decisions on whether to employ virtual reality and extended reality technologies for verification assessments at NASA. In Phase 1, we also developed the preliminary elements of a decision framework for evaluating the use of extended reality technologies in verification assessments. This framework will be further developed in future work but can currently be used to scaffold decisions and discussions surrounding the adoption of XR technologies for design evaluations.
Task Progress:	Before being rated as ready for human-spaceflight, systems must undergo a formal evaluation, known as verification, to ensure that they conform to design requirements. Verifications are conducted with either the product or a high-fidelity mockup standing in for the product. Traditionally, these mockups have been physical (e.g., foam-core and wood), but there is growing interest in exploring new technologies for performing verifications. These technologies include virtual reality (VR), mixed reality (MR), and augmented reality (AR), which are collectively referred to as extended reality (XR) technologies. However, the validity of test results obtained from VR mockups, compared to the current gold standard of physical mockups, remains uncertain. It is unlikely that there is one clean answer as there are many different types of verification outcomes, and each XR technology must be evaluated on its own merits. This is not an issue during developmental testing as the design is still in flux and the total success of the design is not dependent upon the results of a developmental test. Verification assessments, however, only happen once, assuming no change to the design, and the results are used to certify the product. Therefore, establishing the validity of XR mockup-based verification outcomes is essential before considering them for any use in verification tests. Outputs from this investigation include evidence from two studies regarding the validity of VR for a small selection of requirements. Results from the two studies and available literature were also synthesized into a Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis pertaining to the use of XR technologies for verification. And finally, we present a preliminary framework for evaluating XR technologies for use in verifications at NASA.
Bibliography Type:	Description: (Last Updated:)